

Digital Logic Design

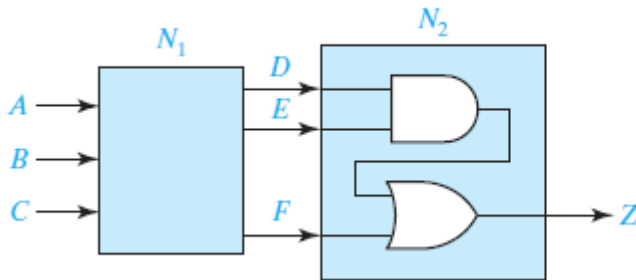
Homework 4

1. Given $F_1 = \sum m(0, 4, 5, 6)$ and $F_2 = \sum m(0, 3, 6, 7)$, find the minterm expression for $F_1 + F_2$. State a general rule for finding the expression for $F_1 + F_2$ given the minterm expansions for F_1 and F_2 . Prove your answer by using the general form of the minterm expansion.
- $F_1 + F_2 = \sum m(0, 3, 4, 5, 6, 7)$
- $F_1 = \sum m(I) \quad I = \{i_1, i_2, \dots, i_p\} \quad F_1 + F_2 = \sum m(I \cup J) \quad F_1 + F_2 = a_0 m_0 + a_3 m_3 + a_4 m_4 + a_5 m_5 + a_6 m_6 + a_7 m_7 = 1$
- $F_2 = \sum m(J) \quad J = \{j_1, j_2, \dots, j_q\}$

2. (a) How many switching functions of two variables (x and y) are there? (b) Give each function in truth table form and in reduced algebraic form.

$2(a) \quad 2^{2^2} = 16 \quad 2(b)$

3. A combinational circuit is divided into two subcircuits N1 and N2 as shown. The truth table for N1 is given. Assume that the input combinations ABC = 110 and ABC = 010 never occur. Change as many of the values of D, E, and F to don't-cares as you can without changing the value of the output Z.



A	B	C	D	E	F	Z
0	0	0	1	1	0	1
0	0	1	0	0	1	1
0	1	0	0	x	x	x
0	1	1	1	1	0	1
1	0	0	1	0	0	0
1	0	1	x	0	1	1
1	1	0	0	x	0	x
1	1	1	0	0	0	0

3.

A	B	C	D	E	F	Z
0	0	0	1	1	0	1
0	0	1	x	x	1	1
0	1	0	x	x	x	x
0	1	1	1	1	0	1
1	0	0	1	0	0	0
1	0	1	x	x	1	1
1	1	0	x	x	x	x
1	1	1	0	0	0	0

4. Work (a) and (b) with the following truth table:

A	B	C	F	G
0	0	0	1	0
0	0	1	x	1
0	1	0	0	x
0	1	1	0	1
1	0	0	0	0
1	0	1	x	1
1	1	0	1	x
1	1	1	1	1

(a) $F = B' + AB$

$G = B'$

- (a) Find the simplest expression for F, and specify the values of the don't-cares that lead to this expression. (b) Repeat (a) for G. (Hint: Can you make G the same as one of the inputs by properly choosing the values for the don't-cares?)

5. Each of three coins has two sides, heads and tails. Represent the heads or tails status of each coin by a logical variable (A for the first coin, B for the second coin, and C for the third) where the logical variable is 1 for heads and 0 for tails. Write a logic function $F(A, B, C)$ which is 1 iff exactly one of the coins is heads

2.(b)

x	1	1	0	0	
y	1	0	1	0	
Z_1	0	0	0	0	0
Z_2	0	0	0	1	$x'y'$
Z_3	0	0	1	0	$x'y$
Z_4	0	0	1	1	$x'y + x'y'$
Z_5	0	1	0	0	xy'
Z_6	0	1	0	1	$xy' + x'y'$
Z_7	0	1	1	0	$xy' + x'y$
Z_8	0	1	1	1	$xy' + x'y + x'y'$
Z_9	1	0	0	0	xy
Z_{10}	1	0	0	1	$xy + x'y'$
Z_{11}	1	0	1	0	$xy + x'y$
Z_{12}	1	0	1	1	$xy + x'y + x'y'$
Z_{13}	1	1	0	0	$xy + xy'$
Z_{14}	1	1	0	1	$xy + xy' + x'y'$
Z_{15}	1	1	1	0	$xy + xy' + x'y$
Z_{16}	1	1	1	1	$xy + xy' + x'y + x'y'$

4.(b) G:

ABC	00	01	11	10
0	0	1	1	x
1	0	1	1	x

C

G=C

5.

A	B	C	outputs	
0	0	0	0	m_0
0	0	1	1	m_1
0	1	0	1	m_2
0	1	1	0	
1	0	0	1	m_4
1	0	1	0	
1	1	0	0	
1	1	1	0	

a) $F = \sum m(1, 2, 4)$

b) $F = \prod M(0, 3, 5, 6, 7)$

b.

N_1		N_2		$N_1 \times N_2$	outputs
A	B	C	D		
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	1	1
0	1	1	0	2	1
0	1	1	1	3	0
1	0	0	0	0	1
1	0	0	1	2	1
1	0	1	0	4	0
1	0	1	1	5	0
1	1	0	0	0	1
1	1	0	1	3	0
1	1	1	0	6	0
1	1	1	1	7	0

a) $F = \sum m(0, 1, 2, 3, 4, 5, 6, 8, 9, 12)$

$$= A'B'C'D' + A'B'C'D + A'B'CD' + A'B'CD + A'BC'D' + A'BC'D + A'BCD' + A'BCD + ABC'D' + ABC'D$$

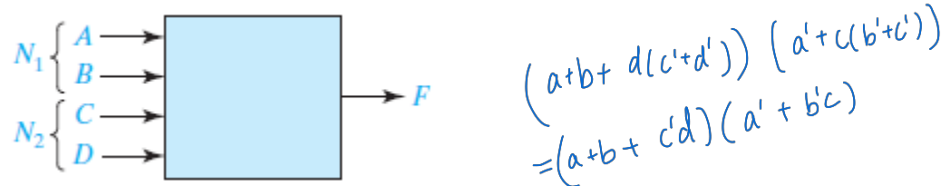
b) $F = \prod M(7, 10, 11, 13, 14, 15)$

$$= (A+B+C+D)(A+B'+C+D')(A+B'+C+D)(A+B+C'+D')(A+B+C+D')(A+B+C+D)$$

after a toss of the coins. Express F (a) as a minterm expansion. (b) as a maxterm expansion.

6. A switching circuit has four inputs as shown. A and B represent the first and second bits of a binary number N_1 . C and D represent the first and second bits of a binary number N_2 . The output is to be 1 only if the product $N_1 \times N_2$ is less than or equal to 2.

(a) Find the minterm expansion for F . (b) Find the maxterm expansion for F . Express your answers in both decimal notation and algebraic form.



7. Given: $F(a,b,c)=abc'+b'$. (a) Express F as a minterm expansion. (Use m-notation.) (b) Express F as a maxterm expansion. (Use M-notation.) (c) Express F' as a minterm expansion. (Use m-notation.) (d) Express F' as a maxterm expansion. (Use M-notation.)
8. Work Problem 4.9 using: $F(a,b,c,d)=(a+b+d)(a'+c)(a'+b'+c')(a+b+c'+d')$.
9. A combinational logic circuit has four inputs (A, B, C , and D) and one output Z . The output is 1 iff the input has three consecutive 0's or three consecutive 1's. For example, if $A = 1, B = 0, C = 0$, and $D = 0$, then $Z = 1$, but if $A = 0, B = 1, C = 0$, and $D = 0$, then $Z = 0$. Design the circuit using one four-input OR gate and four three-input AND gates.
10. Design a combinational logic circuit which has one output Z and a 4-bit input $ABCD$ representing a binary number. Z should be 1 iff the input is at least 5, but is no greater than 11. Use one OR gate (three inputs) and three AND gates (with no more than three inputs each).

7. $F(a,b,c) = abc + b' = b' + bac = b' + ac$
 $x + x'y = x + y$

a	b	c	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

a) $F = \sum m(0,1,4,5,7)$

b) $F = \pi M(2,3,6)$

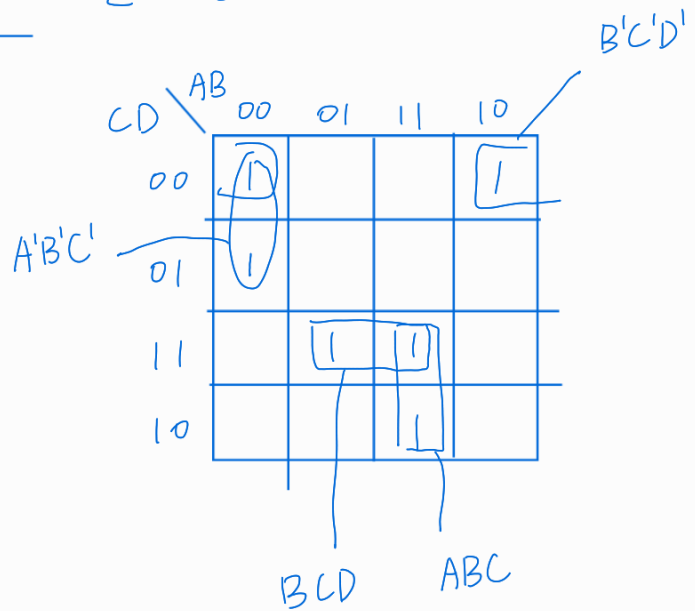
c) $F' = \sum M(2,3,6)$

d) $F' = \pi M(0,1,4,5,7)$

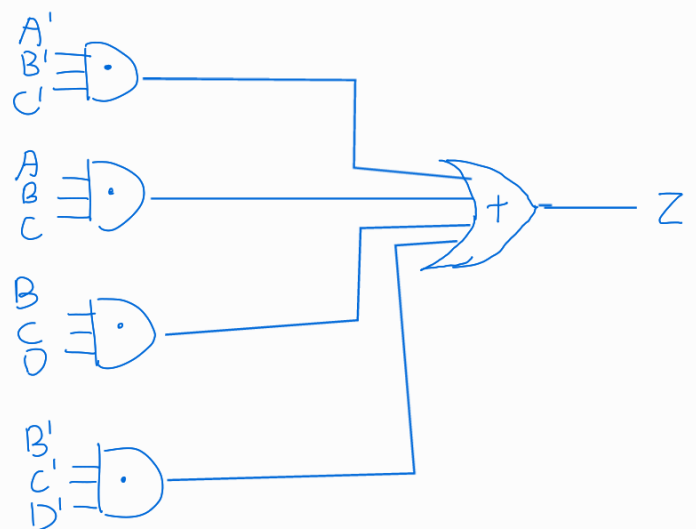
9.

A	B	C	D	Z
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

$Z = \sum(0,1,7,8,14,15)$



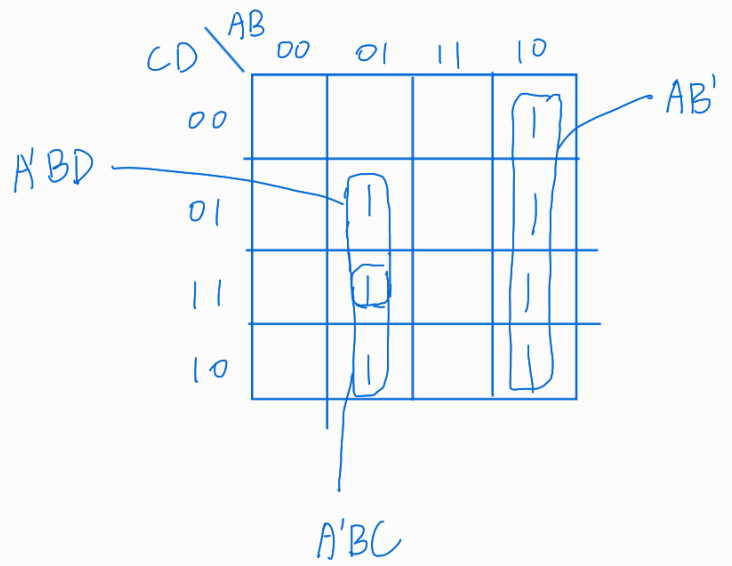
$Z = A'B'C' + B'C'D' + BCD + ABC$



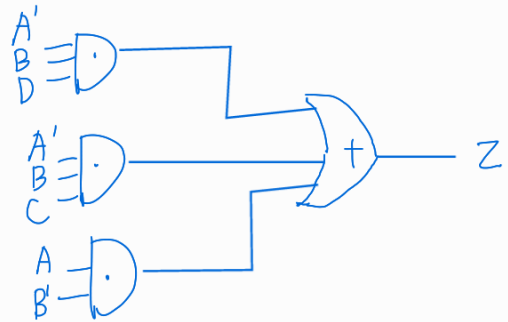
10.

A	B	C	D	Z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$$Z = \Sigma(5, 6, 7, 8, 9, 10, 11)$$



$$Z = A'BD + A'BC + AB'$$



$$8. F(a, b, c, d) = (a+b+d)(a'+c)(a'+b'+c')(a+b+c'+d')$$

$$= (a+b+d(c'+d'))(a'+c(b'+c'))$$

$$= (a+b+c'd)(a'+b'c)$$

a	b	c	d	c'd	a+bc'd	a'	b'	c	a'+b'c	(a+b+c'd)(a'+b'c)
0	0	0	0	0	0	1	1	0	1	0
0	0	0	1	1	1	1	1	0	1	1
0	0	1	0	0	0	1	1	1	1	0
0	0	1	1	0	0	1	1	1	1	0
0	1	0	0	0	1	1	0	0	1	0
0	1	0	1	1	0	1	0	0	1	0
0	1	1	0	0	1	1	0	1	1	1
0	1	1	1	0	1	1	1	0	1	0
1	0	0	0	1	0	0	1	0	0	1
1	0	0	1	0	0	0	1	1	1	1
1	0	1	0	0	1	0	1	1	1	0
1	0	1	1	0	0	0	0	0	0	0
1	1	0	0	1	1	0	0	0	0	0
1	1	0	1	0	1	0	0	1	0	0
1	1	1	0	0	1	0	0	1	0	0
1	1	1	1	0	0	0	0	1	0	0

$$F = \sum m(1, 4, 7, 10, 11)$$

$$F = \prod M(0, 2, 3, 5, 6, 8, 9, 12, 13, 14, 15)$$

$$F' = \sum m(0, 2, 3, 5, 6, 8, 9, 12, 13, 14, 15)$$

$$F' = \prod M(1, 4, 7, 10, 11)$$